

RAJASTHAN IDEATHON · HEALTHCARE INNOVATION

MedScan+

Smart Medicine Safety & Recovery Platform

Verify what you take. Manage how you take it. Recover what you don't.



Medicine mismanagement is a silent, three-part crisis



Wrong Medicine Errors

Prescription is unclear or the wrong strip is picked up at the pharmacy — no one cross-checks it.



Massive Wastage

Most households discard unused medicine — it isn't tracked, returned, or reused safely.



Unaffordable Access

Families don't know where the same medicine is available free or at a fraction of the price nearby.

The gap: No single system today connects what a patient is prescribed, what they actually take, and what happens to the medicine that's left over.

The numbers behind the crisis

3M+

deaths / year worldwide linked to medication errors

WHO, World Patient Safety Day

5.2M

medical injury cases estimated annually in India

Harvard University study

70%

of purchased medicine goes unused or thrown away in 3 of 4 Indian households

LocalCircles national survey

38M

Indians pushed into poverty by medicine spending alone (of 55M total)

PHFI / NSSO-based study

82%

of surveyed households simply threw unused drugs in the dustbin

Bhubaneswar disposal-practice study

₹38,000cr

saved by citizens via Jan Aushadhi in 11 years — proof affordable-access models work

PMBJP, Govt of India (June 2025)

One platform, four connected modules

MedScan+ follows a medicine's full journey — from finding it, to safely using it, to responsibly recovering what's left.



1. FIND

Search a medicine and see the nearest place it's available free (govt/MMNDY) or cheapest (Jan Aushadhi, pharmacy).



2. VERIFY

Scan prescription + medicine packet.
App flags any mismatch in name, strength, or dosage before you take it.



3. MANAGE

Auto-built schedule from the verified prescription — reminders, taken/missed tracking, expiry alerts.



4. RECOVER

Unused, sealed medicine routed to an authorized govt/pharmacy collection point — verified reuse or safe disposal.

Find the medicine, then verify it's the right one



FIND — Locate the medicine nearby

- 1 Search medicine by name (e.g. “Metformin 500mg”)
- 2 App checks 3 sources: Govt/PHC free stock (MMNDY), Jan Aushadhi Kendra, nearby pharmacy
- 3 Results ranked: free → cheapest → nearest, with live map directions
- 4 Reminder to carry prescription for govt/Jan Aushadhi pickup

Sample result: PHC (2.1 km) — FREE | Jan Aushadhi (3.5 km) — ₹8 | Pharmacy (0.8 km) — ₹42



VERIFY — Catch mismatches before harm

- 1 Scan the doctor's prescription (camera / gallery)
- 2 OCR extracts medicine name, strength, dosage, duration
- 3 Scan the medicine packet received from the pharmacy
- 4 App compares both — flags any mismatch instantly



Verified — safe to take



Mismatch — confirm with pharmacist

Stay on schedule, then close the loop responsibly

MANAGE — Auto schedule, zero manual entry

- **Daily Reminders** Notification at the exact dose time set from the verified prescription
- **Taken / Missed Tracking** One-tap logging builds an adherence history over time
- **Expiry Alerts** Warns before a medicine in stock expires unused
- **Refill Alerts** Notifies when stock is about to run out
- **Family Mode** One account tracks medicines for elderly parents & children too

RECOVER — Legally-safe return, not resale

- 1 Eligibility check: sealed? unexpired by 3-6 months?
- 2 Source-based routing: Private → authorized centre; Govt hospital → govt facility; Jan Aushadhi → future integration (pending approval)
- 3 Nearest collection point shown on map
- 4 Pharmacist scans & verifies on drop-off (QR) — creates a digital record
- 5 Eligible → verified reuse/donation; Not eligible → safe scientific disposal

A live dashboard nobody has today

MedScan+ gives Health Department / PHC officials a real-time, block-level view — built entirely from data the app already collects.



Medication Safety

Mismatch alerts by area — spot pharmacy or prescription patterns causing errors



Recovery Volume

How much medicine is collected, where, and its eligible-reuse vs disposal split



Wastage Hotspots

Block-wise wastage pattern to target awareness drives where they're needed most



Policy-Ready Data

Feeds directly into MMNDY planning and any future state health data authority

Timing advantage: Rajasthan already runs Mukhya Mantri Nishulk Dawa Yojana (MMNDY) — spending crores on free medicines with no digital tracking layer today. MedScan+ becomes that missing layer.

Built to be self-sustaining, not donation-dependent

B2G

Govt / PHC / Jan Aushadhi tie-up

Data partnership + recovery-network contract

B2B

Pharmacy chains

Mismatch-check API licensing + sponsored listings in Find

B2B

Insurance companies

Medication-adherence data licensing for risk assessment

B2C

Patients & families

Free core app; premium multi-family tracking subscription

Software-first — low cost, fast to pilot

Why this is realistic to build

- No hardware needed — pure software/app, unlike sensor-based ideas
- Jan Aushadhi's 2,047-medicine list is already public data to integrate
- MMNDY (Rajasthan) provides the free-medicine list via a data partnership
- OCR (prescription/packet scan) uses off-the-shelf, proven technology
- PHC/Kendra infrastructure already exists — no new buildings required

Phased rollout

Phase 1 — Pilot

2–3 PHCs in one block

6 months

Phase 2 — Expand

District-wide, Jan Aushadhi tie-in

1 year

Phase 3 — Scale

State-wide MMNDY integration

2 years

What changes if MedScan+ is built



Fewer preventable harms

from medication mismatches, caught before the dose is taken



Less medicine wasted

sealed, unexpired stock is recovered instead of binned



Lower household cost

families are routed to free/cheapest legitimate sources first



New income for SHGs

local women run and maintain collection-point verification



Real policy data

first-ever block-level visibility for the Health Department



Safer disposal

environmental hazard from dumped medicine drops sharply